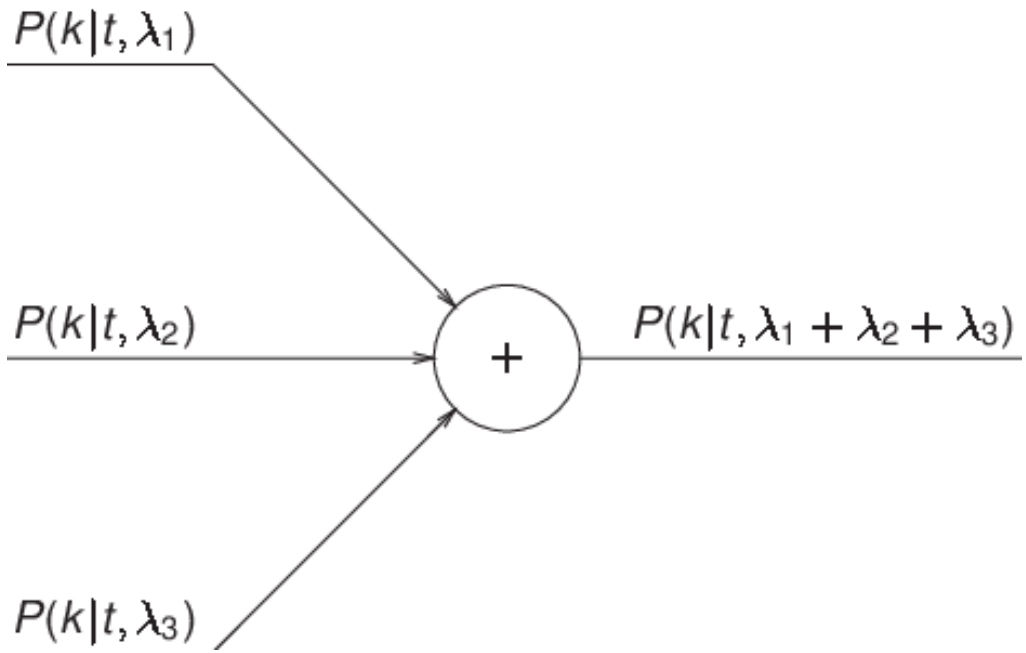
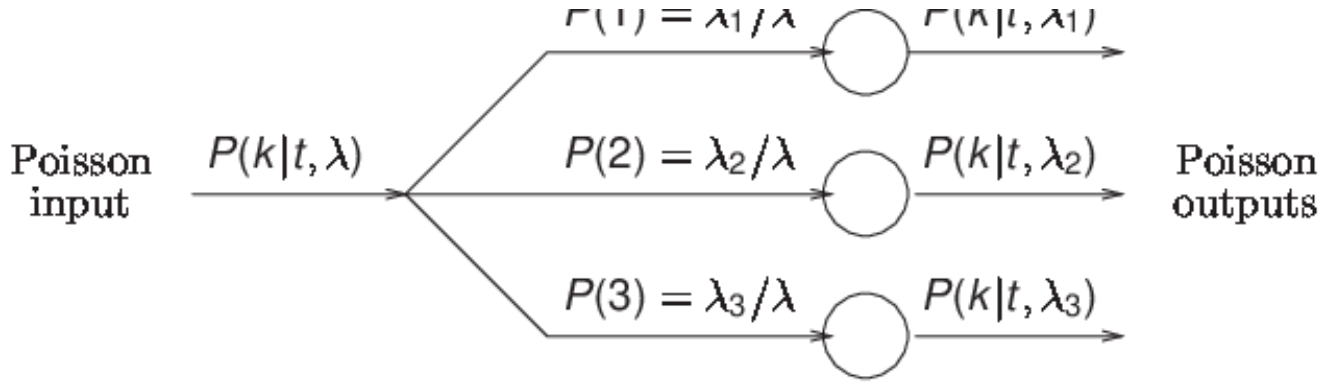
Poisson Processes

2006-09-26



If independent Poisson processes form a single traffic flow overall process is a Poisson with a rate that s the sum of the rates of the separate processes.



If a single Poisson process with rate λ is switched to different

Stochastic Processes

- Also known as probabilistic process, random process
- Can be viewed as a ‘random function’.
- Normal function maps a set of inputs to a point value.

Stochastic Processes (continued)

- A probabilistic process maps a set of inputs to a probability distribution.
- *Wiener process* is the process behind Brownian motion.
- Stock market data is often modelled as a stochastic process.
- Definition: A random process May be defined as a *random function* which, for a particular value, is a random quantity.

Poisson Process

- A simple process which is related to the Poisson distribution: the Poisson process.
- Probabilities of a number of arrivals, k, as a function of time, t, given a 'rate', λ.
-

$$P(k|t,\lambda) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}.$$

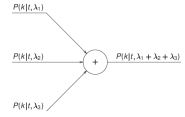
Poisson Process (continued)

- The process is simply the distribution with μ substituted with λt.
- λ is now known as a rate, because product λt gives the expected number of arrivals at time t.
- The mean of the Poisson process (the first moment of k) is given by

$$\langle k \rangle = \lambda t.$$

- Poisson process is a form of counting process where the counts are expected to increase linearly with time.

Properties



sumPoisson

Properties (continued)

- Superposition of Poissons If we combine N independent processes with rates $\lambda_1, \lambda_2, \ldots, \lambda_N$ together the resulting process is also a Poisson process with a combined rate, λ, equal to the sum of each of the input rates: $\lambda = \sum_{n=1}^{N} \lambda_n$. This is shown pictorially in Figure 8.
-

Example of Poisson Superposition

- Example 1, Chapter 4 There are three queues for registration of MSc students from different courses at the University of Sheffield. Students for the first course, Physics and Fine Art, arrive with a rate of 4 students per minute. Students for the second course, The Psychology of Sociology, arrive with a rate of 2 students per minute and students for the third course, Computer Science and Aromatherapy arrive with a rate of 5 students a minute. Assuming the student arrivals are independent and occur as a Poisson process what is the probability that after 15 minutes there have been **a)** 90 student arrivals? **b)**180 arrivals?

Example of Poisson Superposition (continued)

Properties Contd



decomposePoisson

Properties Contd (continued)

- Decomposition of Poisson Flow Assume we have one input channel where arrivals are occurring with a Poisson process with rate λ. This channel then splits into N separate output channels. Each input arrival is then independently assigned to output channel n with probability $P(n)$. In this case the output channels then behave like Poisson processes with rates given by $\lambda_n = P(n)\lambda$. This is shown in Figure 12.
-

Example of Poisson Decomposition

- Example 2, Chapter 4 There is one registration for all MSc courses. Students arrive in the hall and are assigned to registration desks according to the first letter of their surname. Desk 1 ($d = 1$)takes students with names from A-G, desk 2 ($d = 2$) from H-M, desk 3 ($d = 3$) from N to S and desk 4 ($d = 4$) from R to Z. Surnames are distributed amoungst these four categories with the following probability distribution: | d | 1 | 2 | 3 | 4 | | — | — | — | — | — | | $P(d)$ | 0.20 | 0.27 | 0.25 | 0.28 | Assuming the students arrive independently of their surname as a Poisson process at a rate of 10 students per minute what is the probability that **a)**after 5 minutes 5 students have arrived at desk 1? **b)** after 10 minutes 20 students have arrived at desk 4?

Example of Poisson Decomposition (continued)

Properties III

- Independence of Arrivals Consider a single Poisson process, if we look within this process at two *disjoint* time intervals (*i.e.* periods of time that don't overlap). Assume we have k_1 arrivals in the first interval and k_2 arrivals in the second interval. The distribution of arrivals in the two intervals are independent of each other, $P(k_1, k_2) = P(k_1) P(k_2)$.

Properties IV

- Similarity of Arrival Distributions for Identical Intervals
  Consider a single Poisson process with rate λ, let the number of arrivals between time s and $s + t$ be given by k_1. Now consider a second independent Poisson process with the same rate λ, let the number of arrivals between time r and $r + t$ be given by k_2. The distribution of k_1 and k_2 are then equal,

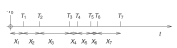
$$P(k_1) = P(k_2).$$

Interarrival Times and Poisson Processes

- Interarrival time: time between arrivals.
- Think of buses.
- X_i, are continuous values, what is their PDF?.

Interarrival Distribution

- Example 3, Chapter 4. Derivation of Interarrival Time Distribution In the figure above we show different arrival times $T_1, \ldots, T_n$ for the process. You can think of these as the arrival times of the buses. We are interested in computing the distribution of the variables $X_i = T_i - T_{i-1}$: the times between bus arrivals. In this example you will derive this PDF for the interarrival times. [1]



interArrivals

[1] The interarrival times, X_i, are continuous values, so they will be governed by a probability density function rather than a discrete probability distribution.

Interarrival Distribution (continued)

Properties of the Exponential Distribution

- The Memoryless property

$$P(X > s + t | X > t) = P(X > s)$$

- In English, the probability of the waiting time being greater than $s + t$ given that the waiting time is already greater than t is the same as the probability of the waiting time being greater than s.
- Mean and Variance

$$p(x) = \lambda e^{-\lambda x}$$

The mean of an exponential can be found to be,

$$\langle x \rangle_{p(x)} = \frac{1}{\lambda}$$

and the variance is

Thanks!

- company: Trent AI
- book: The Atomic Human
- twitter: @lawrennd
- The Atomic Human
- newspaper: Guardian Profile Page
- blog: http://inverseprobability.com

Further Reading

- Sections on the Poisson process and the exponential distribution of Bertsekas and Gallager (1992)

Exercises

- Tutorial Sheet 1: Poisson probabilities, exponential sampling, and the Poisson variance of Lawrence (2006)

References

Bertsekas, D.P., Gallager, R.G., 1992. Data networks, Second. ed. Prentice-Hall, Upper Saddle River, NJ.

Lawrence, N.D., 2006. Network performance analysis tutorial 1.